

Sheth L.U.J. And Sir M.V. College

Aim:- Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CLI MODE:-

```
Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)
File Edit Format Run Options Window Help
print("=" * 55)
print("      DIFFIE-HELLMAN KEY EXCHANGE")
print("=" * 55)

p = 23
g = 15

print("\nPublic Parameters:")
print("Prime Number (p):", p)
print("Generator (g):", g)

alice_private = int(input("\nEnter Alice's Private Key: "))
bob_private = int(input("Enter Bob's Private Key: "))

alice_public = pow(g, alice_private, p)
bob_public = pow(g, bob_private, p)

print("\n--- Public Key Generation ---")
print("Alice's Public Key:", alice_public)
print("Bob's Public Key:", bob_public)

alice_shared = pow(bob_public, alice_private, p)
bob_shared = pow(alice_public, bob_private, p)

print("\n--- Shared Secret Calculation ---")
```

```
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print("\n--- Shared Secret Calculation ---")
print("Alice's Shared Secret:", alice_shared)
print("Bob's Shared Secret:", bob_shared)

if alice_shared == bob_shared:
    print("\n✓ KEY EXCHANGE SUCCESSFUL")
    print("Both Alice and Bob have the same shared secret.")
else:
    print("\nX KEY EXCHANGE FAILED")
    print("Shared secrets do not match.")

print("\n--- FAILED VERIFICATION TEST ---")
test_secret = bob_shared + 1

print("Alice's Shared Secret:", alice_shared)
print("Modified Bob's Secret:", test_secret)

if alice_shared == test_secret:
    print("\n✓ KEY EXCHANGE SUCCESSFUL")
    print("Both shared secrets match.")
else:
    print("\nX KEY EXCHANGE FAILED")
    print("Shared secrets do not match.")

print("=" * 55)
```

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Output:-

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

=====
DIFFIE-HELLMAN KEY EXCHANGE
=====

Public Parameters:
Prime Number (p): 23
Generator (g): 15

Enter Alice's Private Key: 25
Enter Bob's Private Key: 50

--- Public Key Generation ---
Alice's Public Key: 17
Bob's Public Key: 13

--- Shared Secret Calculation ---
Alice's Shared Secret: 12
Bob's Shared Secret: 12

✓ KEY EXCHANGE SUCCESSFUL
Both Alice and Bob have the same shared secret.

--- FAILED VERIFICATION TEST ---
Alice's Shared Secret: 12
Modified Bob's Secret: 13
✗ KEY EXCHANGE FAILED
Shared secrets do not match.
=====

>>>
```

GUI MODE:-

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
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import tkinter as tk
from tkinter import messagebox

def calculate_exchange():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())

        alice_private = int(alice_private_entry.get())
        bob_private = int(bob_private_entry.get())

        if p <= 1 or g <= 0:
            messagebox.showwarning(
                "Invalid Input",
                "Please enter valid values for p and g."
            )
            return

        if alice_private <= 0 or bob_private <= 0:
            messagebox.showwarning(
                "Invalid Private Key",
                "Private keys must be positive numbers."
            )
            return

        alice_public = pow(g, alice_private, p)
```

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```
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alice_public = pow(g, alice_private, p)
bob_public = pow(g, bob_private, p)

alice_shared = pow(
    bob_public,
    alice_private,
    p
)

bob_shared = pow(
    alice_public,
    bob_private,
    p
)

alice_public_entry.config(state="normal")
alice_public_entry.delete(0, tk.END)
alice_public_entry.insert(0, str(alice_public))
alice_public_entry.config(state="readonly")

bob_public_entry.config(state="normal")
bob_public_entry.delete(0, tk.END)
bob_public_entry.insert(0, str(bob_public))
bob_public_entry.config(state="readonly")

Ln: 62 Col: 0
28°C Partly sunny Search 09:00 10-08-2026
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
File Edit Format Run Options Window Help

alice_shared_entry.config(state="normal")
alice_shared_entry.delete(0, tk.END)
alice_shared_entry.insert(0, str(alice_shared))
alice_shared_entry.config(state="readonly")

bob_shared_entry.config(state="normal")
bob_shared_entry.delete(0, tk.END)
bob_shared_entry.insert(0, str(bob_shared))
bob_shared_entry.config(state="readonly")

global original_alice_shared
global original_bob_shared

original_alice_shared = alice_shared
original_bob_shared = bob_shared

if alice_shared == bob_shared:

    status_title.config(
        text="KEY EXCHANGE SUCCESSFUL",
        fg="#16803c"
    )

    status_message.config(
        text="Alice and Bob have generated the same shared secret.",
        fg="#4b5563"
    )

Ln: 62 Col: 0
28°C Partly sunny Search 09:00 10-08-2026
```

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```
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    status_icon.config(
        text="✓",
        fg="#16803c"
    )

except ValueError:

    messagebox.showerror(
        "Invalid Input",
        "Please enter numbers only."
    )

def failed_test():

    try:

        alice_secret = int(alice_shared_entry.get())
        bob_secret = int(bob_shared_entry.get())

        if not alice_secret or not bob_secret:
            messagebox.showwarning(
                "No Key Available",
                "First generate the shared key."
            )
        return
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvuc/Documents/Practical No. 5 Cyber Security.py (3.14.6)
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    modified_bob_secret = bob_secret + 1

    bob_shared_entry.config(state="normal")
    bob_shared_entry.delete(0, tk.END)
    bob_shared_entry.insert(0, str(modified_bob_secret))
    bob_shared_entry.config(state="readonly")

    if alice_secret == modified_bob_secret:

        status_icon.config(
            text="✓",
            fg="#16803c"
        )

        status_title.config(
            text="KEY EXCHANGE SUCCESSFUL",
            fg="#16803c"
        )

        status_message.config(
            text="The shared secrets match.",
            fg="#4b5563"
        )

    else:

        status_icon.config(
            text="X",
```

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```
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    fg="#c62828"
)

status_title.config(
    text="KEY EXCHANGE FAILED",
    fg="#c62828"
)

status_message.config(
    text="Shared secrets do not match. Tampering detected.",
    fg="#4b5563"
)

except ValueError:

    messagebox.showwarning(
        "Verification Error",
        "Generate the shared key first."
    )

def clear_all():

    alice_private_entry.delete(0, tk.END)
    bob_private_entry.delete(0, tk.END)

    alice_public_entry.config(state="normal")
    alice_public_entry.delete(0, tk.END)

Ln: 62 Col: 0
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
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    alice_public_entry.config(state="readonly")

    bob_public_entry.config(state="normal")
    bob_public_entry.delete(0, tk.END)
    bob_public_entry.config(state="readonly")

    alice_shared_entry.config(state="normal")
    alice_shared_entry.delete(0, tk.END)
    alice_shared_entry.config(state="readonly")

    bob_shared_entry.config(state="normal")
    bob_shared_entry.delete(0, tk.END)
    bob_shared_entry.config(state="readonly")

    status_icon.config(
        text="●",
        fg="#7c3aed"
    )

    status_title.config(
        text="READY FOR KEY EXCHANGE",
        fg="#7c3aed"
    )

    status_message.config(
        text="Enter the private keys and generate the shared secret.",
        fg="#4b5563"
    )

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```

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```
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root = tk.Tk()

root.title("Diffie-Hellman Key Exchange")

root.geometry("900x700")

root.configure(bg="#f5f3ff")

root.resizable(False, False)

header = tk.Frame(
    root,
    bg="ffffff",
    height=110
)

header.pack(fill="x")

header.pack_propagate(False)

title = tk.Label(
    header,
    text="DIFFIE-HELLMAN",
    font=("Segoe UI", 25, "bold"),
    bg="ffffff",
    fg="#4c1d95"
)

Ln: 294 Col: 0
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)
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title.pack(pady=(18, 0))

subtitle = tk.Label(
    header,
    text="Secure Key Exchange Simulator",
    font=("Segoe UI", 11),
    bg="ffffff",
    fg="#6b7280"
)

subtitle.pack(pady=5)

main = tk.Frame(
    root,
    bg="#f5f3ff"
)

main.pack(
    fill="both",
    expand=True,
    padx=35,
    pady=25
)

Ln: 294 Col: 0
```

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```
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parameter_box = tk.Frame(
    main,
    bg="#ede9fe",
    padx=20,
    pady=15
)

parameter_box.pack(fill="x")

parameter_title = tk.Label(
    parameter_box,
    text="PUBLIC PARAMETERS",
    font=("Segoe UI", 10, "bold"),
    bg="#ede9fe",
    fg="#6d28d9"
)

parameter_title.grid(
    row=0,
    column=0,
    columnspan=4,
    sticky="w",
    pady=(0, 8)
)

tk.Label(
    parameter_box,
```

```
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    text="Prime Number (p)",
    font=("Segoe UI", 10),
    bg="#ede9fe",
    fg="#374151"
).grid(row=1, column=0, padx=10)

p_entry = tk.Entry(
    parameter_box,
    width=12,
    font=("Segoe UI", 11),
    bg="ffffff",
    fg="#111827",
    relief="flat",
    justify="center"
)

p_entry.grid(
    row=1,
    column=1,
    padx=10
)

p_entry.insert(0, "23")

tk.Label(
    parameter_box,
    text="Generator (g)",
```

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```
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    font=("Segoe UI", 10),
    bg="#ede9fe",
    fg="#374151"
).grid(row=1, column=2, padx=10)

g_entry = tk.Entry(
    parameter_box,
    width=12,
    font=("Segoe UI", 11),
    bg="ffffff",
    fg="#111827",
    relief="flat",
    justify="center"
)

g_entry.grid(
    row=1,
    column=3,
    padx=10
)

g_entry.insert(0, "5")

people_frame = tk.Frame(
    main,
```

```
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    bg="#f5f3ff"
)

people_frame.pack(
    fill="x",
    pady=20
)

alice_box = tk.Frame(
    people_frame,
    bg="ffffff",
    highlightbackground="#c4b5fd",
    highlightthickness=2,
    padx=20,
    pady=15
)

alice_box.grid(
    row=0,
    column=0,
    sticky="nsew"
)

tk.Label(
    alice_box,
```

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```
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text="ALICE",
font=("Segoe UI", 16, "bold"),
bg="#ffffff",
fg="#7c3aed"
).pack()

tk.Label(
    alice_box,
    text="Sender / Participant A",
    font=("Segoe UI", 9),
    bg="#ffffff",
    fg="#9ca3af",
).pack(pady=(0, 15))

tk.Label(
    alice_box,
    text="Private Key",
    font=("Segoe UI", 9, "bold"),
    bg="#ffffff",
    fg="#4b5563",
).pack(anchor="w")

alice_private_entry = tk.Entry(
    alice_box,
    font=("Segoe UI", 11),
    bg="#f9fafb",
    fg="#111827",
    relief="flat",
    justify="center"
)

alice_private_entry.pack(
    fill="x",
    ipady=7,
    pady=(4, 12)
)

tk.Label(
    alice_box,
    text="Public Key",
    font=("Segoe UI", 9, "bold"),
    bg="#ffffff",
    fg="#4b5563",
).pack(anchor="w")

alice_public_entry = tk.Entry(
    alice_box,
    font=("Segoe UI", 11),
    bg="#f3f4f6",
    fg="#7c3aed",
    relief="flat",
    justify="center"
)


```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
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fg="#111827",
relief="flat",
justify="center"
)

alice_private_entry.pack(
    fill="x",
    ipady=7,
    pady=(4, 12)
)

tk.Label(
    alice_box,
    text="Public Key",
    font=("Segoe UI", 9, "bold"),
    bg="#ffffff",
    fg="#4b5563",
).pack(anchor="w")

alice_public_entry = tk.Entry(
    alice_box,
    font=("Segoe UI", 11),
    bg="#f3f4f6",
    fg="#7c3aed",
    relief="flat",
    justify="center"
)


```

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```
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alice_public_entry.pack(
    fill="x",
    ipady=7,
    pady=(4, 12)
)

alice_public_entry.config(state="readonly")

tk.Label(
    alice_box,
    text="Shared Secret",
    font=("Segoe UI", 9, "bold"),
    bg="ffffff",
    fg="#4b5563"
).pack(anchor="w")

alice_shared_entry = tk.Entry(
    alice_box,
    font=("Segoe UI", 12, "bold"),
    bg="#f3f4f6",
    fg="#16803c",
    relief="flat",
    justify="center"
)

alice_shared_entry.pack(
    fill="x",
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
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    ipady=7,
    pady=(4, 0)
)

alice_shared_entry.config(state="readonly")

channel = tk.Frame(
    people_frame,
    bg="#f5f3ff",
    width=130
)

channel.grid(
    row=0,
    column=1,
    padx=10
)|

channel.grid_propagate(False)

tk.Label(
    channel,
    text="e",
    font=("Segoe UI", 32, "bold"),
    bg="#f5f3ff",
```

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```
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    fg="#f59e0b"
).pack(pady=(45, 0))

tk.Label(
    channel,
    text="PUBLIC\nCHANNEL",
    font=("Segoe UI", 8, "bold"),
    bg="#f5f3ff",
    fg="#9ca3af",
).pack()

bob_box = tk.Frame(
    people_frame,
    bg="ffffff",
    highlightbackground="#fdba74",
    highlightthickness=2,
    padx=20,
    pady=15
)

bob_box.grid(
    row=0,
    column=2,
    sticky="nsew"
)
```

Ln: 551 Col: 1

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
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tk.Label(
    bob_box,
    text="BOB",
    font=("Segoe UI", 16, "bold"),
    bg="ffffff",
    fg="#ea580c"
).pack()

tk.Label(
    bob_box,
    text="Receiver / Participant B",
    font=("Segoe UI", 9),
    bg="ffffff",
    fg="#9ca3af",
).pack(pady=(0, 15))

tk.Label(
    bob_box,
    text="Private Key",
    font=("Segoe UI", 9, "bold"),
    bg="ffffff",
    fg="#4b5563",
).pack(anchor="w")

bob_private_entry = tk.Entry(
    bob_box,
```

Ln: 595 Col: 0

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```
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font=("Segoe UI", 11),
bg="#f9fafb",
fg="#111827",
relief="flat",
justify="center"
)

bob_private_entry.pack(
    fill="x",
    ipady=7,
    pady=(4, 12)
)

tk.Label(
    bob_box,
    text="Public Key",
    font=("Segoe UI", 9, "bold"),
    bg="ffffff",
    fg="#4b5563"
).pack(anchor="w")

bob_public_entry = tk.Entry(
    bob_box,
    font=("Segoe UI", 11),
    bg="#f3f4f6",
    fg="#ea580c",
    relief="flat",
    justify="center"
)
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
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    justify="center"
)

bob_public_entry.pack(
    fill="x",
    ipady=7,
    pady=(4, 12)
)

bob_public_entry.config(state="readonly")

tk.Label(
    bob_box,
    text="Shared Secret",
    font=("Segoe UI", 9, "bold"),
    bg="ffffff",
    fg="#4b5563"
).pack(anchor="w")

bob_shared_entry = tk.Entry(
    bob_box,
    font=("Segoe UI", 12, "bold"),
    bg="#f3f4f6",
    fg="#16803c",
    relief="flat",
    justify="center"
)
```

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```
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bob_shared_entry.pack(
    fill="x",
    ipady=7,
    pady=(4, 0)
)

bob_shared_entry.config(state="readonly")

buttons = tk.Frame(
    main,
    bg="#f5f3ff"
)

buttons.pack(pady=5)

generate_button = tk.Button(
    buttons,
    text="🔑 Generate Shared Key",
    command=calculate_exchange,
    font=("Segoe UI", 10, "bold"),
    bg="#7c3aed",
    fg="white",
    activebackground="#6d28d9",
    activeforeground="white",
    relief="flat",

```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)
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    cursor="hand2",
    padx=20,
    pady=11
)

generate_button.grid(
    row=0,
    column=0,
    padx=6
)

failed_button = tk.Button(
    buttons,
    text="⚠️ Test Failed Verification",
    command=failed_test,
    font=("Segoe UI", 10, "bold"),
    bg="#f59e0b",
    fg="white",
    activebackground="#d97706",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    padx=20,
    pady=11
)

failed_button.grid(
    row=0,

```

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```
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    row=0,
    column=1,
    padx=6
)

clear_button = tk.Button(
    buttons,
    text="0 Reset",
    command=clear_all,
    font=("Segoe UI", 10, "bold"),
    bg="#6b7280",
    fg="white",
    activebackground="#4b5563",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    padx=20,
    pady=11
)

clear_button.grid(
    row=0,
    column=2,
    padx=6
)
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
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status_box = tk.Frame(
    main,
    bg="ffffff",
    highlightbackground="ddd6fe",
    highlightthickness=1,
    padx=20,
    pady=12
)

status_box.pack(
    fill="x",
    pady=(15, 0)
)

status_icon = tk.Label(
    status_box,
    text="●",
    font=("Segoe UI", 18, "bold"),
    bg="ffffff",
    fg="#7c3aed"
)

status_icon.pack()

status_title = tk.Label(
    status_box,
    text="READY FOR KEY EXCHANGE",

```

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```
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    status_box,
    text="READY FOR KEY EXCHANGE",
    font=("Segoe UI", 11, "bold"),
    bg="ffffff",
    fg="#7c3aed"
)

status_title.pack()

status_message = tk.Label(
    status_box,
    text="Enter the private keys and generate the shared secret.",
    font=("Segoe UI", 9),
    bg="ffffff",
    fg="#4b5563"
)

status_message.pack(pady=(3, 0))

footer = tk.Label(
    root,
    text="Diffie-Hellman • Secure Key Agreement • Educational Demonstration",
    font=("Segoe UI", 9),
    bg="#f5f3ff",
    fg="#9ca3af"
)

Ln: 595 Col: 0
28°C Partly sunny
```

```
*Practical No. 5 Cyber Security.py - C:/Users/mvluc/Documents/Practical No. 5 Cyber Security.py (3.14.6)*
File Edit Format Run Options Window Help

status_title.pack()

status_message = tk.Label(
    status_box,
    text="Enter the private keys and generate the shared secret.",
    font=("Segoe UI", 9),
    bg="ffffff",
    fg="#4b5563"
)

status_message.pack(pady=(3, 0))

footer = tk.Label(
    root,
    text="Diffie-Hellman • Secure Key Agreement • Educational Demonstration",
    font=("Segoe UI", 9),
    bg="#f5f3ff",
    fg="#9ca3af"
)

footer.pack(pady=(0, 12))

root.mainloop()

Ln: 595 Col: 0
28°C Partly sunny
```

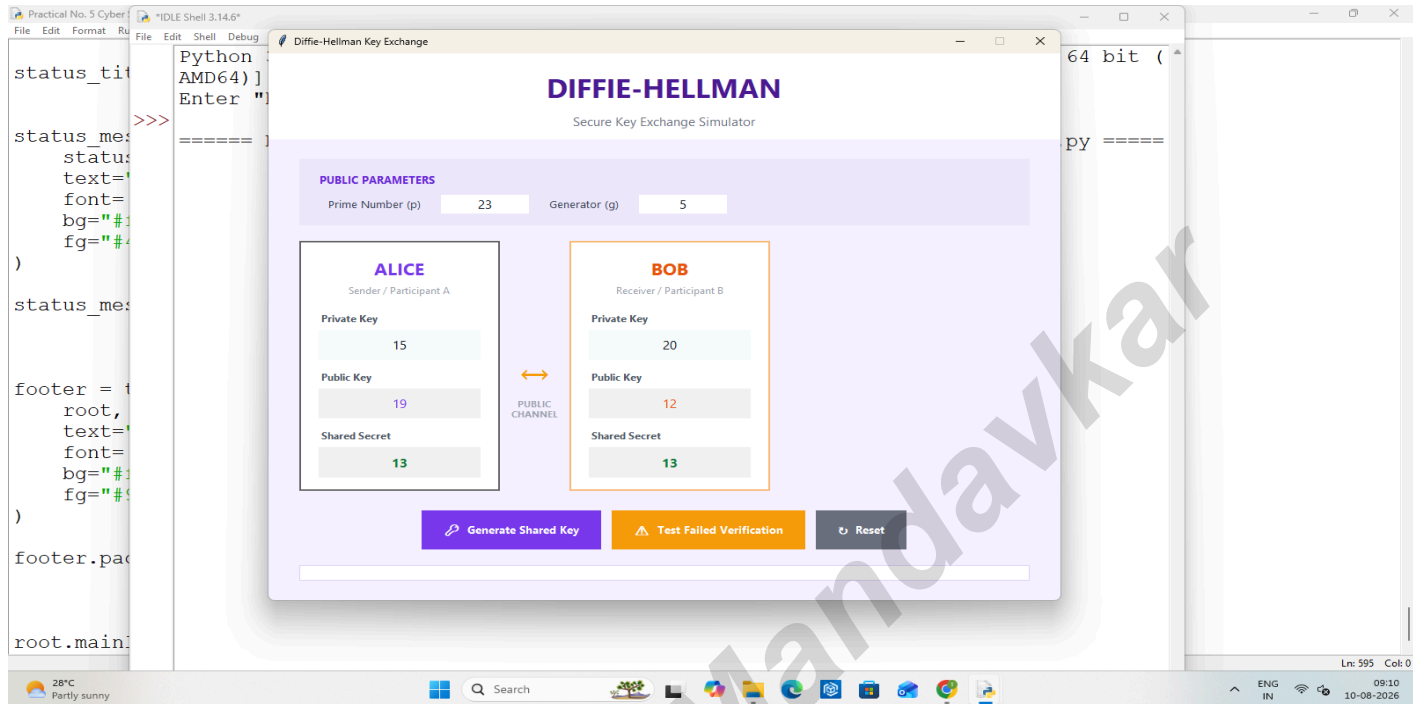
Name:- Chetan Mandavkar

Roll No. T093

Subject:- Cyber And Information Security

Sheth L.U.J. And Sir M.V. College

Output:-



Name:- Chetan Mandavkar

Roll No. T093

Subject:- Cyber And Information Security